Quantum Mechanics III

Exercise sheets

One sheet accompanies each lecture. This booklet contains questions and selected study hints. Try a problem before reading its hints. If you have the separate *Worked answers*, try the questions before consulting it. The course website publishes the questions, lecture examples, study hints and self-check answers; it does not publish that worked-answer booklet. These are computations and applications; the course's general proofs are in the lectures. Unless specified, inner products are antilinear in the first slot and tensor factors are in the order stated. Allow roughly 30–75 minutes per sheet, with the longer computations marked. This is a planning allocation, not an observed completion time. Problems marked optional develop additional applications and are not needed to follow the subsequent lectures.

Contents

1 States and Gibbs operators 3

2 Preparations and complex ensemble freedom 4

3 Effect probabilities and three paths 5

4 Conjugation, ring currents and phase choices 6

5 Time covariance and sector states 7

6 Parts and squeezed correlations 8

7 Choi matrices and dilations 9

8 Relaxation and cavity attenuation 10

9 Unsharp joint readings 11

10 Pointer noise, finite disturbance and Zeno 12

11 Environmental overlap and rates 13

12 Global tests and retained records 14

13 Context values, locality and trajectories 15

14 Collapse noise and relative states 16

Sheet 1

States and Gibbs operators

1. **Thermal spin.** For $H = (\hbar\nu/2)\sigma_z$, $\nu > 0$, compute $Z(\beta)$, both energy probabilities, $\langle H \rangle$ and $\operatorname{Var} H$. Check the zero- and high-temperature limits.
2. **A particle in a box.** The energies are $\varepsilon_n = \hbar^2\pi^2 n^2/(2mL^2)$, $n \geq 1$. Show the Gibbs partition sum is finite for $\beta > 0$ by comparison with a geometric series. Give the probabilities and the zero-temperature limit. No closed form for the sum is required.
3. **Why the free particle differs (longer).** In momentum space its Gibbs operator is multiplication by $f(p) = e^{-\beta p^2/(2m)}$. Choose $p_0 > 0$ so $f \geq c > 0$ on $[-p_0, p_0]$. For any finite-rank operator T, find a unit vector supported in that interval with $Tu = 0$ and deduce $\|f(P) - T\| \geq c$ (here $f(P)$ denotes this momentum multiplication). Conclude that this Gibbs operator is not compact and hence not a positive trace-class operator. Use Lecture 1's proved compactness implication, not a divergent spatial trace formula.

> **First hint**
>
> Choose more independent interval-supported vectors than the rank of the finite-rank operator. Their images all lie in its finite-dimensional range.

> **Further hint**
>
> Take a nontrivial linear dependence of those images and use the same coefficients on the input vectors. Normalise the resulting nonzero kernel vector, then apply the positive lower bound on the multiplier over its support.

Sheet 2

Preparations and complex ensemble freedom

1. For $\rho = \mathrm{diag}(4/5, 1/5)$, compute purity and the variance of $E = |0\rangle\langle 0|$. Verify that the equally weighted states $\psi_\pm = \sqrt{4/5}\,|0\rangle \pm \sqrt{1/5}\,|1\rangle$ give the same density operator. Are these two states orthogonal?
2. Construct a label-space unitary in the lecture's convention $x_i = \sum_j u_{ij} y_j$, where $x_0 = |0\rangle/\sqrt{2}$, $x_1 = |1\rangle/\sqrt{2}$ and $y_\pm = (|0\rangle \pm i\,|1\rangle)/2$. Check both the relation and $uu^\dagger = \mathbb{1}$. Also construct Problem 1's own label unitary, relating $x_0 = \sqrt{4/5}\,|0\rangle$, $x_1 = \sqrt{1/5}\,|1\rangle$ to $y_\pm = \psi_\pm/\sqrt{2}$. Why can no physical unitary map $|0\rangle, |1\rangle$ to that nonorthogonal pair?
3. A coherent amplitude has mean number $\mu = 3/4$. Write the first three probabilities after averaging its phase uniformly. Compute mean number and vacuum probability. Why would averaging only an overall vector phase leave the original pure density operator unchanged?

Sheet 3

Effect probabilities and three paths

1. For $\rho = (\mathbb{1} + \sigma_x/3 + \sigma_z/2)/2$ and $F = (\mathbb{1} + (2/5)\sigma_x)/2$, check positivity of the state and effect, then compute $\mathrm{Tr}(\rho F)$ and the mean of $2F$. Is $2F$ itself an effect?
2. For the qubit projection assignment $h(\mathbf{n}) = (1+n_z^5)/2$, verify the antipodal rule. Values on the coordinate axes would force which Bloch vector for a density representation? Test that representation at $n_z = 1/\sqrt{2}$. Explain which premise of Busch's theorem is absent.
3. In $\mathbb{C}^3$ prepare $(e_A + e_B + e_C)/\sqrt{3}$, keep evolution fixed at the identity and detect $(e_A + e^{i\phi}e_B + e_C)/\sqrt{3}$. Compute all seven nonempty-path probabilities per original trial, without renormalising filtered states. Verify $I_3 = 0$ for arbitrary ϕ, and list the probabilities at $\phi = \pi$.

Sheet 4

Conjugation, ring currents and phase choices

1. Let C be conjugation in the σ_z basis. Compute $C\sigma_j C^{-1}$ and its action on a Bloch vector. Check directly that $C(i\psi) = -iC\psi$ and that ray transition probabilities are preserved.
2. On the ring take $\theta = \pi/3$. Compute the three levels for $k = -1, 0, 1$ in units $\hbar^2\pi^2/(2mL^2)$ and the ground current with $\theta = -q\Phi/\hbar$. Identify their time-reversed partners. At $\theta = \pi$, give the two ground-branch currents and explain why no unique derivative of the ground energy specifies the current there.
3. If U_t is a genuine unitary group, define $\mathcal{V}_t = e^{i\alpha t^3}U_t$, where α is real and has units time^{-3}. Compute its multiplier and give a rephasing restoring the group law. Does strong continuity of this family alone make it a group?

Sheet 5

Time covariance and sector states

1. For $H = \mathrm{diag}(0, \Delta, 2\Delta)$, $\Delta > 0$, compute the return probability of $(|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$. Replace U_t by $e^{ict}U_t$ and find the new Hamiltonian and its domain. Which part of the probability changes?
2. **Wrapped energy shifts.** On $L^2(0, L)$ put $H = \alpha Q$, $T = -P_\theta/\alpha$, $\alpha > 0$, with $\psi(x + L) = e^{i\theta}\psi(x)$ for the twisted periodic extension. For $0 < s < \alpha L$ compute $W_s = e^{isT/\hbar}$ from the momentum eigenfunctions, then compute $W_s H W_s^{-1}$ as a multiplication operator. Locate the interval where it differs from $H - s\mathbb{1}$. Which covariance condition fails despite $[T, H]\psi = i\hbar\psi$ on compactly supported smooth vectors?

   > **First hint**
   >
   > Apply the exponential to one twisted momentum eigenfunction. Compare the resulting phase with evaluating that eigenfunction at a translated argument. Keep the twisted extension when the argument leaves the interval.

   > **Further hint**
   >
   > Split the interval according to whether the translated argument wraps through an endpoint. The twist phases cancel in conjugating the multiplication operator, but its coordinate multiplier changes by one interval length on the wrapped piece. Compare this piece with the proposed energy shift.

3. Let u be a unit vector in the integer sector and v one in the half-integer sector, and put $\psi = \frac{1}{2}u + i\frac{\sqrt{3}}{2}v$. Find its unique block-diagonal representative on the two-block algebra, its expectation of $2\mathbb{1}_{\mathrm{int}} \oplus \mathbb{1}_{\mathrm{half}}$, and its restricted purity status. Construct the phase-matched cross-sector witness and compute its mean in the ray and block mixture. Is that witness an admitted observable?

Sheet 6

Parts and squeezed correlations

1. Let (e_j) and (f_k) be orthonormal bases of the two factors. For $\Psi = (e_0 \otimes f_0 + i e_1 \otimes f_1)/\sqrt{2}$ and $\rho = (2/3)\,|\Psi\rangle\langle\Psi| + (1/3)\,|e_2 \otimes f_0\rangle\langle e_2 \otimes f_0|$, compute both reduced states and their purities. Is the composite pure?
2. **Squeezed Gaussian (longer).** Let $s > 0$ and $r \geq 0$. Normalise

$$\psi_r \propto \exp\left[-\frac{e^{2r}(x_1 - x_2)^2}{4s^2} - \frac{e^{-2r}(x_1 + x_2)^2}{4s^2}\right].$$

   Use $u = (x_1 - x_2)/\sqrt{2}$, $v = (x_1 + x_2)/\sqrt{2}$ to compute the two collective variances and the local position and momentum variances. Check the local uncertainty product and evaluate the lecture's witness.

   > **First hint**
   >
   > Use the squared modulus for normalisation and position moments. The orthogonal change of coordinates has Jacobian of magnitude one, and it turns the density into a product of two one-dimensional Gaussians.

   > **Further hint**
   >
   > Write each local coordinate as a sum or difference of the new coordinates. Their cross moment vanishes. For momentum moments, apply the momentum derivatives to the wavefunction, rather than differentiating its squared modulus. Check the unsqueezed case before interpreting the collective result.

3. Independent classical shifts add variance n_x to each local position and n_p to each local momentum. Find the resulting collective witness. At $r = (\log 2)/2$, compare $n_x/s^2 = s^2 n_p/\hbar^2 = 0.1$ and 0.3. Does failure of this sufficient test prove separability?

Sheet 7

Choi matrices and dilations

1. For the decay channel $K_0 = \mathrm{diag}(\sqrt{1-\gamma}, 1)$, $K_1 = \sqrt{\gamma}\,|1\rangle\langle 0|$, $0 \le \gamma \le 1$, compute the unnormalised Choi matrix in the input-reference/output order $00, 01, 10, 11$. Find its eigenvalues and check its trace and output partial trace.
2. At $\gamma = 3/4$, write the isometry J and a complete unitary matrix on system plus a ready qubit that implements it. Use basis $00, 01, 10, 11$ and ready environment $|0\rangle$. Check orthonormal columns.
3. Compare the two Bell preparations $\Phi_\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}$. Find their marginals, then apply the CNOT with system controlling environment. Compute the final system marginals. What information, absent from either initial marginal, determines their different futures?

Sheet 8

Relaxation and cavity attenuation

1. Solve the lecture's Bloch equations from the excited state and from $(|0\rangle + e^{i\pi/3}|1\rangle)/\sqrt{2}$. Express the excited population and the $+x$ reading probability in terms of T_1, T_2, ν_0. Which measurement separates the two times?
2. A unital map contracts the Bloch components by (b, b, a), with $a = 0.6$, $b = 0.9$. Check positivity on isolated states, find the smallest Choi eigenvalue and its eigenvector, and the negative expectation after acting on half of $\Omega/\sqrt{2}$. Is it a channel?
3. **Vacuum cavity attenuation (longer).** A two-mode isometry is specified on number states by

$$J_\eta |n\rangle = \sum_{\ell=0}^{n} \sqrt{\binom{n}{\ell}} \eta^{(n-\ell)/2} (1-\eta)^{\ell/2} |n-\ell\rangle \otimes |\ell\rangle, \qquad 0 \le \eta \le 1.$$

   Use the endpoint convention $0^0 = 1$. Check its orthonormal images, then apply it to the coherent-state series. Find the reduced output and its mean number. With $\eta = e^{-\kappa t}$ and free phase rotation at angular frequency ν, find $\alpha(t)$. This computes the explicit attenuation channel; no general theorem about an unbounded Lindblad generator is requested.

> **First hint**
>
> Two images of different number states belong to orthogonal total-number subspaces. For the norm of one image, use the binomial theorem.

> **Further hint**
>
> After substituting the coherent-state series, replace the input number by the sum of the two output numbers. Simplify the factorials before separating the double series. Identify each normalised output factor before taking the partial trace.

Sheet 9

Unsharp joint readings

1. Let $\mathbf{n}_j = (\cos(2\pi j/3), \sin(2\pi j/3), 0)$ for $j = 0, 1, 2$. Show that $F_j = (\mathbb{1} + \mathbf{n}_j \cdot \boldsymbol{\sigma})/3$ is a POVM. Compute its square roots and Naimark isometry on $|0\rangle$; list the three probabilities. These are a concrete dilation computation, not another proof of the general theorem.
2. For $s, t = \pm 1$, test the proposed joint effects $M_{st} = [\mathbb{1} + \eta(s\sigma_x + t\sigma_z)]/4$, $0 \le \eta \le 1$. Compute spectra and marginals. For which η is this a joint POVM? Does the positivity test of this particular construction alone prove a necessity theorem about every possible joint reading?
3. **Phase-space matrix elements and moments (longer).** For $n, m \in \{0, 1, 2\}$ compute $\pi^{-1} \int_{\mathbb{C}} \langle h_n, \psi_\alpha\rangle\langle\psi_\alpha, h_m\rangle \, d^2\alpha$ from the coherent-state series, using polar coordinates. You may use $\int_0^\infty e^{-u} u^n \, du = n!$. This checks nine matrix elements, not the full resolution theorem, which is cited in L9. For coherent input $\beta = (1 + i)/\sqrt{2}$, find the mean physical position and momentum and their observed variances under the vacuum-generated phase-space POVM. Compare total and added spread products. State the parameter-port and ideal-detector assumptions of the eight-port implementation.

> **First hint**
>
> Use polar coordinates in the coherent-state coefficients. Integrate the angular phase first to distinguish diagonal from off-diagonal matrix elements.

> **Further hint**
>
> For a diagonal element, substitute the squared radius into the remaining radial integral and keep the polar Jacobian. For the physical moments, use the calibration factors in Lecture 9 and separate the input coherent-state spread from the extra spread of the reading before comparing their products.

Sheet 10

Pointer noise, finite disturbance and Zeno

1. For the centred chirped meter $\varphi_b(y) = (2\pi\sigma_M^2)^{-1/4} e^{-(1-ib)y^2/(4\sigma_M^2)}$, with $b \in \mathbb{R}$ and $\sigma_M > 0$, compute its position and momentum spreads, then ϵ_Q, η_P for the lecture's impulsive coupling with $g > 0$. When is the product $\hbar/2$? If a linear phase $e^{ip_0 y/\hbar}$ is added, which mean system kick must be tracked separately?
2. **Finite noise–disturbance calculation (optional).** Use a CNOT with system controlling a meter initially $|0\rangle$, and system initially $|+y\rangle$. Put $A = \sigma_z$, $B = \sigma_x$, pointer $Z_M = \sigma_z$, and define $N = U^\dagger(\mathbb{1}\otimes Z_M)U - A\otimes\mathbb{1}$, $D = U^\dagger(B\otimes\mathbb{1})U - B\otimes\mathbb{1}$. Compute $\epsilon_A = \sqrt{\langle N^2\rangle}$, $\eta_B = \sqrt{\langle D^2\rangle}$, the input spreads and $|\langle[A,B]\rangle|/2$. Compare both $\epsilon_A\eta_B$ and $\epsilon_A\eta_B + \epsilon_A\Delta B + \Delta A\eta_B$ to that commutator quantity. This evaluates the finite Ozawa quantities, not a derivation of a universal inequality.
3. **Repeated selected tests (optional).** A qubit starts in $|0\rangle$ and evolves with $H = (\hbar\nu_R/2)\sigma_x$. After each of n equal intervals over time t, perform an ideal projective test and select outcome 0. Compute the probability of the full selected sequence and its limit as $n \to \infty$. Which experimental assumptions are essential?

Sheet 11

Environmental overlap and rates

1. The controlled-environment overlap is $c = 0.3{+}0.4i$, and the input qubit is $+x$. Compute the reduced density matrix, its Bloch vector, purity and $+x$ probability. Can an unread local test tell whether this state arose from an entangled pure composite or a proper preparation mixture?
2. With $\Lambda_{\mathrm{dec}} = 10^{12}\,\mathrm{m^{-2}s^{-1}}$, evaluate the Gaussian kernel multiplier across $d = 1\,\mu\mathrm{m}$ after one second and the corresponding e^{-1} time. For an input with finite momentum variance, find the added momentum variance. Explain why a mathematically valid channel does not by itself justify this quadratic rate for a resolved gas collision.
3. Double the diameter of the lecture's air grain to $20\,\mu\mathrm{m}$, keeping the same bath and resolved separation. Estimate the rate and time. For the separate thermal-photon molecule model, keep its coefficient $10^{10}\,\mathrm{m^{-2}s^{-1}}$ and take separation 20 nm. Find its time. If only its radius doubles within the same Rayleigh model, what factor changes its rate at fixed separation?

Sheet 12

Global tests and retained records

1. For $\Psi_\phi = (|000\rangle + e^{i\phi}\,|111\rangle)/\sqrt{2}$ compute the one- and two-qubit marginals. Find the means of $B_x = |000\rangle\langle 111| + |111\rangle\langle 000|$ and $B_y = -i\,|000\rangle\langle 111| + i\,|111\rangle\langle 000|$. Give a phase-matched witness with mean one for every ϕ, against zero for the incoherent mixture.
2. In the two-dimensional span of calibrated orthogonal outputs χ_0, χ_1, form the equal superposition χ. Compute the eigenvalues of $|\chi\rangle\langle\chi| - |\chi_0\rangle\langle\chi_0|$ and its trace norm. Compare with the robust theorem's separation at $\eta = 0.1$ and $\eta = 1/2$. Why must calibration explicitly place each output in its labelled set S_i? This is a finite application of the cited theorem, not its general proof.
3. Copy the three-qubit chain's last record to a fourth qubit, producing $(|0000\rangle + |1111\rangle)/\sqrt{2}$. Undo only the two original CNOTs, leaving the fourth record untouched. Compute the final system $+x$ probability. Starting afresh from the four-qubit state, in which order must all three couplings be reversed to restore the original interference? Why is undoing the third-to-fourth coupling ineffective after its control record has already been reset?

Sheet 13

Context values, locality and trajectories

1. Evaluate the six signed context terms and χ for the all-+1 assignment to the Peres–Mermin square, then for the assignment with only $v(A_{11}) = -1$. Which quantum context product does the first assignment fail? What extra independence assumption is needed when averaging assignments sampled in different experimental contexts?
2. A toy response model has $\lambda = \pm 1$ with equal weights, settings $x, y \in \{0, 1\}$, and deterministic outcomes $\alpha = \lambda$, $\beta = \lambda(-1)^{xy}$. Check PI, OI and averaged no-signalling. Compute its CHSH value and compare with the quantum upper value $2\sqrt{2}$ recalled in QM-I. This toy is not asserted to be a quantum or Bohmian realisation.
3. **A non-vacuous guiding flow (longer).** A freely spreading Gaussian is

$$\psi_t(x) = (2\pi\sigma^2)^{-1/4}(1 + it/\tau)^{-1/2} e^{-x^2/[4\sigma^2(1+it/\tau)]}, \qquad \tau = 2m\sigma^2/\hbar.$$

   Compute its velocity, solve $\dot{X}_t = v_t(X_t)$, $X_0(a) = a$, and verify transport of the Gaussian position density. Does any trajectory cross zero? Which global-existence difficulties excluded by this example still need the cited general theorem for other states?

> **First hint**
>
> Differentiate the logarithm of the packet with respect to position. Only its imaginary part contributes to the guiding velocity. Separate the real and imaginary parts of the complex width before solving the trajectory equation.

> **Further hint**
>
> The velocity is linear in position, so integrate the logarithmic change of a nonzero trajectory's absolute value and then handle the zero trajectory separately. For density transport, compare the Gaussian width and the Jacobian of the trajectory map. This example is nowhere zero; list which hypotheses that makes easy rather than generalising to packets with nodes.

Sheet 14

Collapse noise and relative states

1. A labelled rigid pointer has $N = 10^{23}$, $\lambda_{\mathrm{GRW}} = 10^{-16}\,\mathrm{s}^{-1}$, $r_C = 10^{-7}\,\mathrm{m}$ and separation $8r_C$. Compute the hit and damping rates and first-hit mean waiting time. For the support model take $\ell = 0.02r_C$ and bound the Gaussian tail using $\mathrm{erfc}(a) \le e^{-a^2}/(a\sqrt{\pi})$ for $a > 0$ (given). Apply the lecture's bound at $\epsilon = 0.01$. What changes if separation is only $0.1r_C$? Distinguish the rate calculation from branch selection.
2. **GRW kinetic-energy injection.** The one-hit random kick has zero mean and variance $\hbar^2/(2r_C^2)$ per Cartesian component. For a free particle of finite initial kinetic energy, compute its one-hit mean-energy change and the 1D and 3D rates at hit rate λ_{GRW}. Sum for N equal labelled masses, then estimate the 3D power using the parameters above, $m = 1.67 \cdot 10^{-27}\,\mathrm{kg}$ and $\hbar = 1.055 \cdot 10^{-34}\,\mathrm{J\,s}$. Why is this not directly a temperature rise or a current experimental exclusion claim?
3. In the capstone use input $\rho = \mathbb{1}_2/2$ and purification $\Omega = (|0_S 0_R\rangle + |1_S 1_R\rangle)/\sqrt{2}$. Write the four branch vectors using E_j, the $\pi/3$ rotation and M_k of L10. Compute their weights. For each projected branch, what operator remains if only the two probes are traced out, and what further trace gives the system relative state?